

Subject: Final Project Proposal – Numerov Method for Quantum Bound States

I would like to propose a final project focused on solving the one-dimensional time-independent Schrödinger equation using the Numerov method.

The goal of the project is to implement a numerical solver for bound states in various potentials by combining the Numerov integration scheme with a shooting method. The boundary value problem will be reformulated as a root-finding problem for a mismatch in boundary conditions, allowing eigenvalues to be computed using bracketing and bisection. The project will focus on symmetric potentials, allowing the use of parity (even/odd states) to simplify the boundary conditions. The correctness of the solver will be verified through comparison with analytical solutions where available and through convergence behavior.

The main objectives are:

- Implement the Numerov method for second-order differential equations.
- Develop a shooting method with bisection to determine eigenvalues.
- Solve for bound states in several potentials (core focus):
 - Infinite square well (validation against exact solutions)
 - Harmonic oscillator (validation against exact solutions)
 - Quartic double-well potential (tunneling and energy splitting)
 - Finite square well (additional nontrivial test case)
 - Finite rectangular barrier and double-barrier potentials (exploration of scattering behavior such as transmission and resonances)
- Normalize and analyze the resulting wavefunctions.
- Perform convergence studies with respect to grid spacing and domain size to verify numerical accuracy.
- Compare the Numerov method with a fourth-order Runge–Kutta (RK4) integrator for the harmonic oscillator to analyze accuracy, efficiency, and method suitability for Schrödinger-type problems.
- Visualize the root-finding process by plotting the mismatch function as a function of trial energy.
- Analyze physical behavior such as parity, node structure, and tunneling-induced energy splitting.

The project will include:

- A modular Python implementation (Numerov solver, shooting method, potentials).
- Visualization of wavefunctions and probability densities.
- Convergence plots demonstrating numerical accuracy.
- Parameter studies (e.g., double-well barrier dependence).
- Comparison with analytical solutions where available.

- Validation of scattering results through comparison with known analytical transmission coefficients.
- Method comparison plots (Numerov vs RK4) highlighting differences in convergence behavior.
- A short report summarizing methods, results, and numerical accuracy.

This project connects directly to numerical ODE methods and eigenvalue problems discussed in the course, particularly the use of shooting methods for boundary value problems, convergence analysis, root-finding techniques, and the comparison of specialized and general-purpose integrators (Numerov vs Runge–Kutta). It also extends to scattering problems as a secondary component, illustrating transmission, reflection, and resonance phenomena within the same numerical framework.

Best regards,
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